

Watermelon juice: potential functional drink for sore muscle relief in athletes.

l-Citrulline is an excellent candidate to reduce muscle soreness, and watermelon is a fruit rich in this amino acid. This study investigated the potential of watermelon juice as a functional drink for athletes. An in vitro study of intestinal absorption of l-citrulline in Caco-2 cells was performed using unpasteurized (NW), pasteurized (80 °C for 40 s) watermelon juice (PW) and, as control, a standard of l-citrulline. l-citrulline bioavailability was greater when it was contained in a matrix of watermelon and when no heat treatment was applied. In the in vivo experiment (maximum effort test in a cycloergometer), seven athletes were supplied with 500 mL of natural watermelon juice (1.17 g of l-citrulline), enriched watermelon juice (4.83 g of l-citrulline plus 1.17 g from watermelon), and placebo. Both watermelon juices helped to reduce the recovery heart rate and muscle soreness after 24 h.